

SIH26126 — Mission Report

Recorded run: run_1080p · goal UTM 44N E 403562.52 N 1450024.12 (map 56.00, 34.00) · no GNSS in the loop

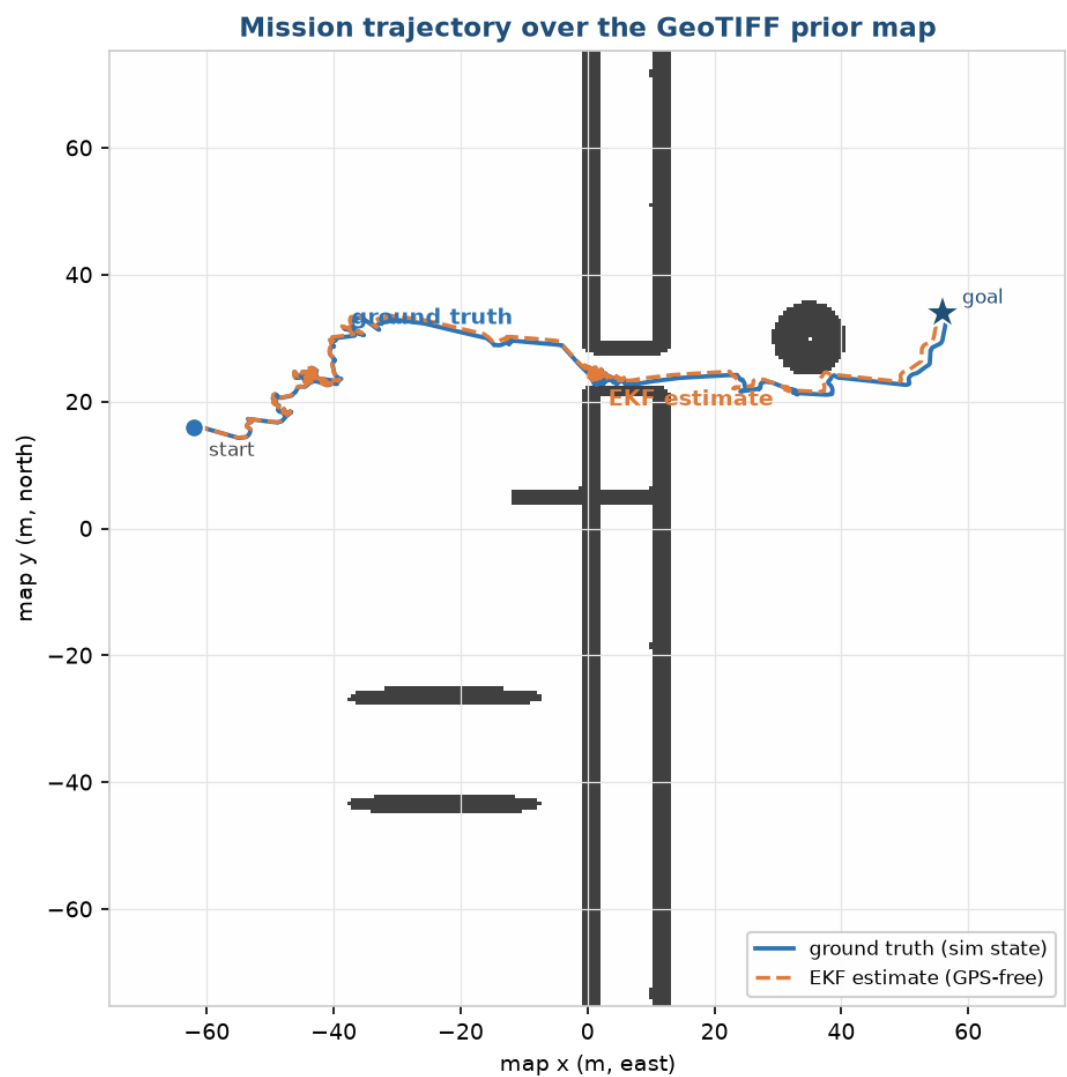
MISSION SUCCESSFUL — GOAL REACHED

1. Result summary

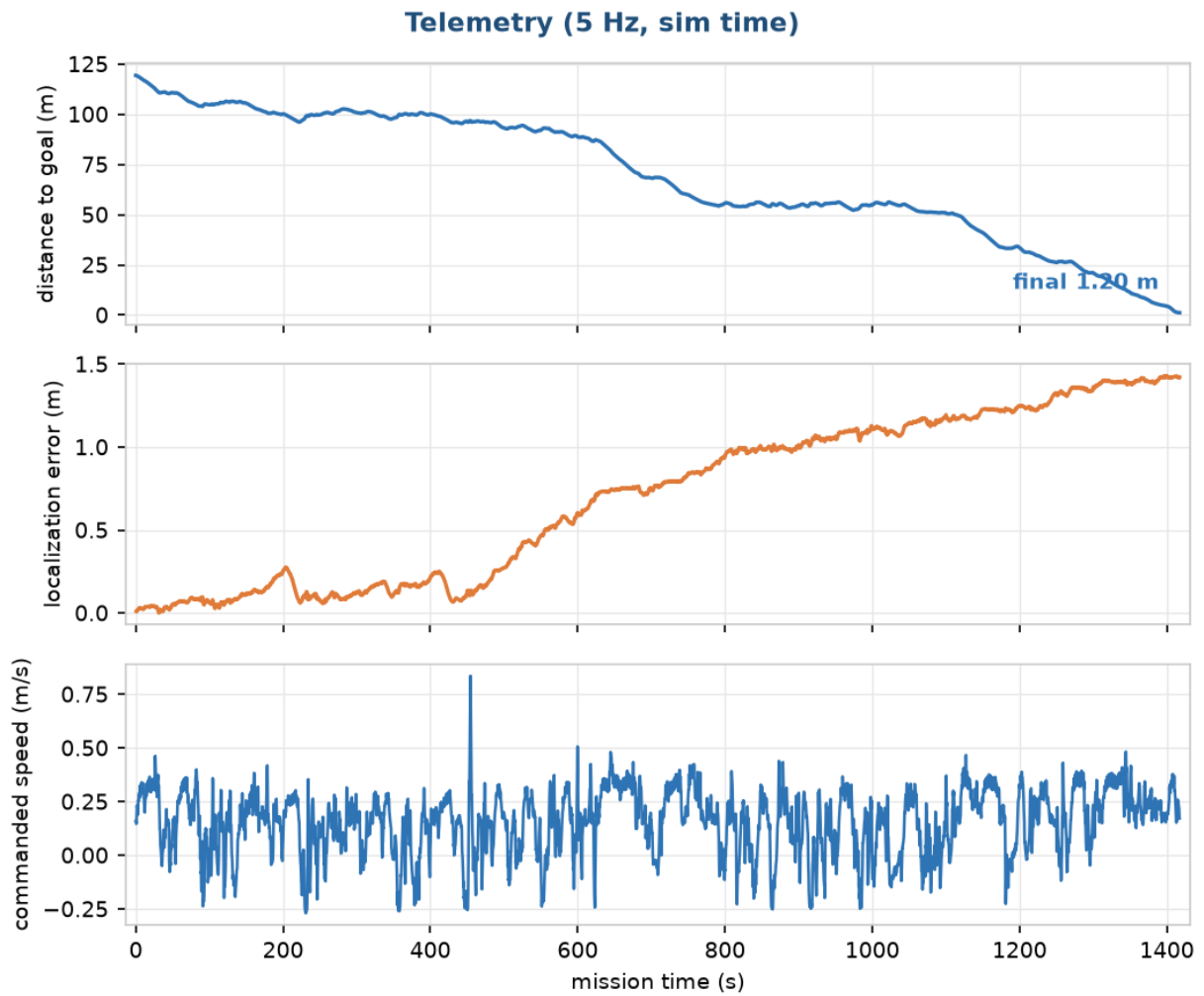
Quantity	Value	Source
Initial distance to goal	119.33 m	ground truth at first log row
Final distance to goal	1.20 m	ground truth
Localization error at goal	1.42 m	ground truth – EKF — the GPS-free claim
Mean / max localization error	nan / nan m	whole mission
Path length driven	295.17 m	integrated ground truth
Mission duration	1417 s sim time (23.6 min)	telemetry log
Time in motion	1222 s	cmd_v > 0.05 m/s
Average moving speed	0.24 m/s	path / moving time
Recovery manoeuvres	1	Nav2 feedback
Nav2 final status	4 (SUCCEEDED)	NavigateToPose result

Ground truth comes from the simulator state and is used for scoring only — nothing in the navigation stack subscribes to it. The EKF position is what the rover actually believed and navigated on, computed from body-frame velocities and IMU heading alone.

2. Trajectory

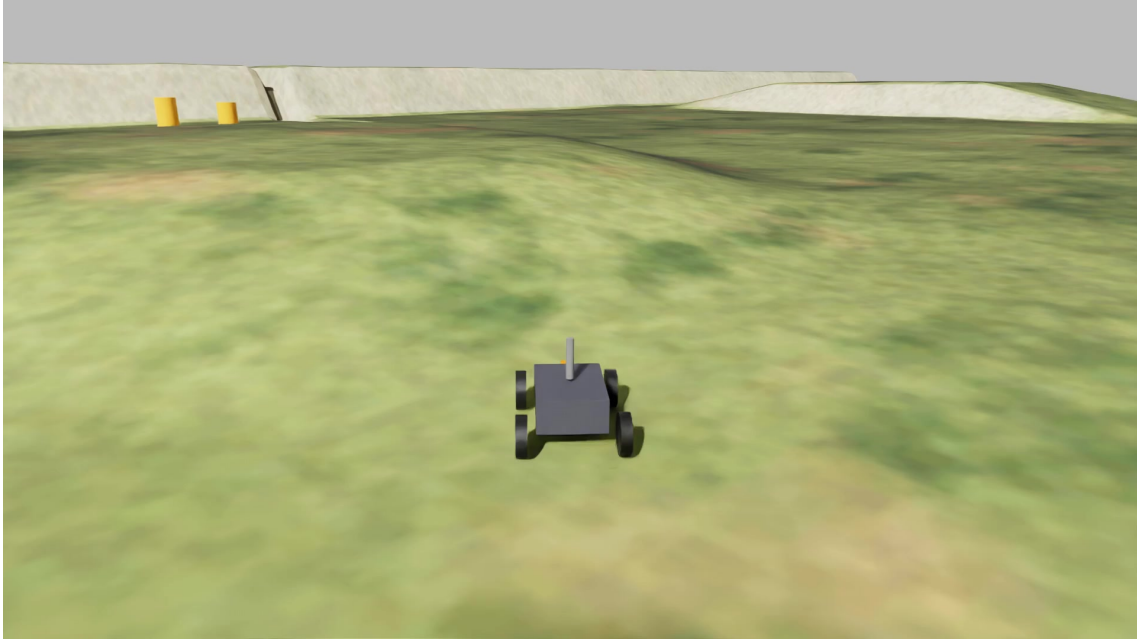


3. Telemetry

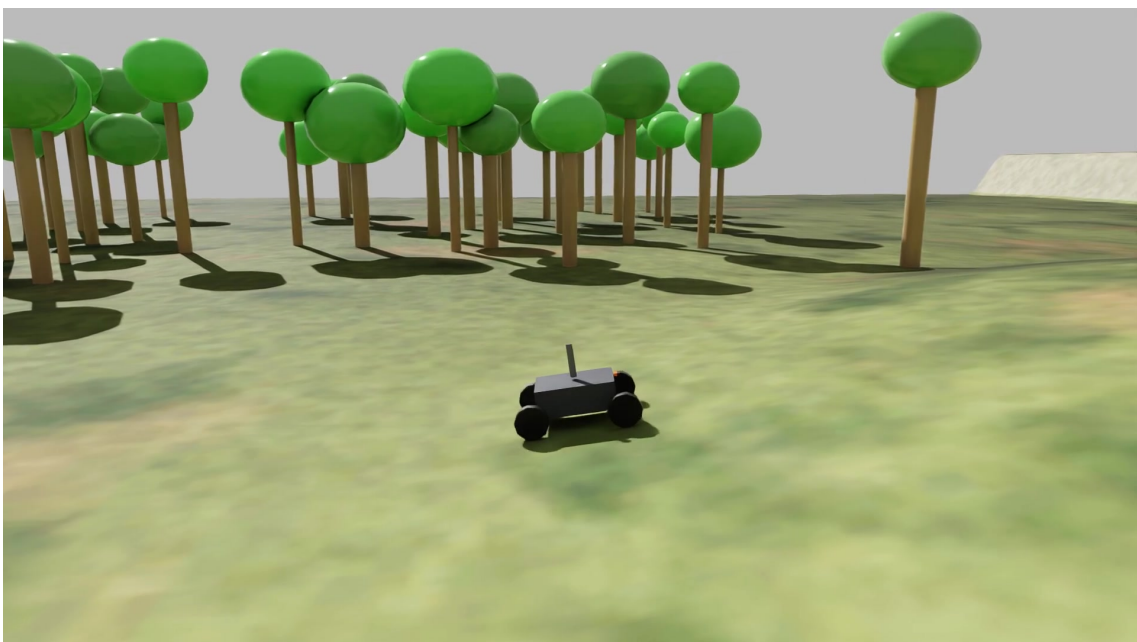


Flat stretches in distance-to-goal with nonzero speed are the tunnel and detours around discovered obstacles; drops to zero speed mark recovery manoeuvres.

4. Camera perspectives (from the recordings)



behind the rover — 1920×1080

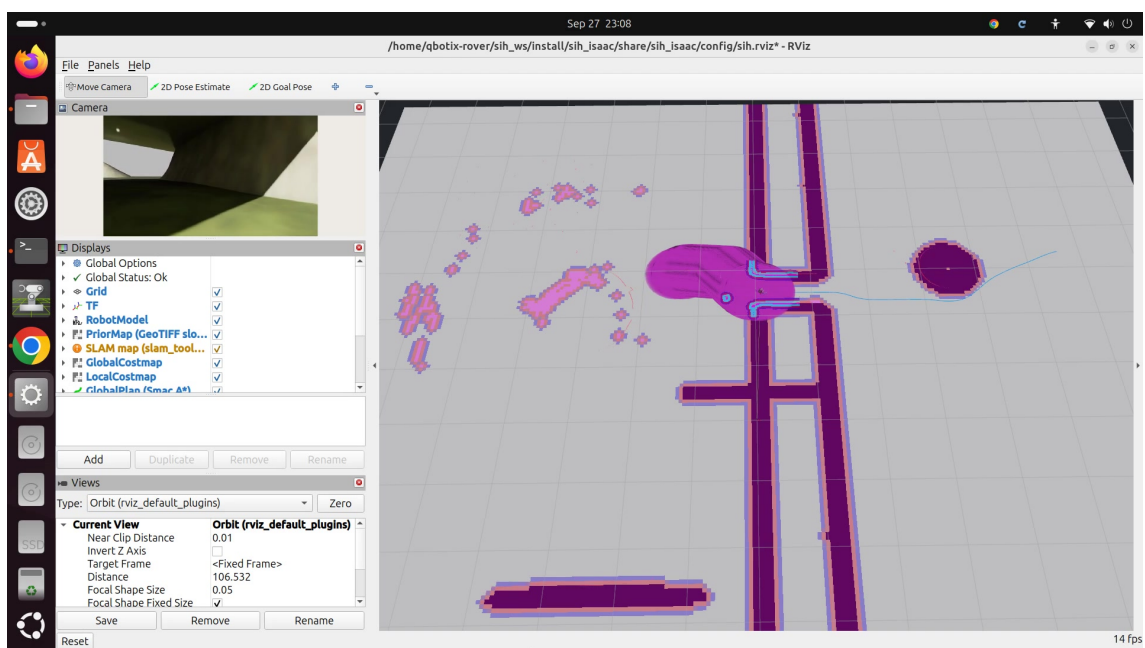


side angle — 1920×1080

4. Camera perspectives (continued)



top-down — 1920×1080



RViz (the robot's belief) — 1920×1080

5. Recording inventory

File	Resolution	Duration	Size
isaac_behind.mp4	1920×1080	1432 s	181.8 MB
isaac_side.mp4	1920×1080	1432 s	201.4 MB
isaac_top.mp4	1920×1080	1432 s	95.8 MB
rviz.mp4	1920×1080	1836 s	51.2 MB
QSLAM_highlight.mp4 (highlight reel)	1920×1080	90 s	40.5 MB

All files are in /home/qbotix-rover/sih_ws/mission_recordings/run_1080p. The three isaac_*.mp4 views are rendered in-simulation at 15 fps of sim time (behind / side follow cameras and a heading-stabilised top-down view); rviz.mp4 is a screen capture of the navigation stack's live belief — costmaps, plans, lidar and TF.